Topic: The logistic equation

Question: A population of insects grows exponentially with a growth constant $k = 7$. How long will it take for the population to triple, assuming time t is measured in hours?

Answer choices:

A 0.649 hours

B 0.157 hours

C 2.333 hours

D 0.429 hours

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Solution: B

If we say that P_0 is the original population, and $3P_0$ is triple the original population, then the exponential growth equation is

$$3P_0 = P_0 e^{7t}$$

$$\frac{3P_0}{P_0} = e^{7t}$$

$$3 = e^{7t}$$

Apply the natural log to both sides of the equation, canceling the $\ln e$, and then rearrange.

$$\ln 3 = \ln e^{7t}$$

$$\ln 3 = 7t$$

$$t = \frac{\ln 3}{7}$$

$t \approx 0.157$ hours

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Topic: The logistic equation

Question: A population of flies doubled in 5 hours. Assuming exponential growth, how long would it take for the population to increase by 10-fold?

Answer choices:

A 2.41 hours

B 3.32 hours

C 16.61 hours

D 1.51 hours

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Solution: C

We've been told that the population doubles in 5 hours. If we call its original size P_0, then we can say that it grows to $2P_0$ in $t = 5$ hours. Substituting these values into the exponential growth equation gives

$$2P_0 = P_0 e^{k(5)}$$

$$\frac{2P_0}{P_0} = e^{k(5)}$$

$$2 = e^{5k}$$

To solve for k, we'll apply the natural log to both sides of the equation, cancelling the $\ln e$, and then rearrange.

$$\ln 2 = \ln e^{5k}$$

$$\ln 2 = 5k$$

$$k = \frac{\ln 2}{5}$$

Now that we have a value for the growth constant k, we can figure out how long it would take for the population to increase 10 fold. If we say that P_0 is the original population, and $10P_0$ is the resulting population that's 10 times larger, then the exponential equation is

$$10P_0 = P_0 e^{\frac{\ln 2}{5}t}$$

$$\frac{10P_0}{P_0} = e^{\frac{\ln 2}{5}t}$$

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$$10 = e^{\frac{\ln 2}{5}t}$$

Apply the natural log to both sides of the equation, canceling the $\ln e$, and then rearrange.

$$\ln 10 = \ln e^{\frac{\ln 2}{5}t}$$

$$\ln 10 = \frac{\ln 2}{5}t$$

$$t = \frac{5 \ln 10}{\ln 2}$$

$t \approx 16.61$ hours

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Topic: The logistic equation

Question: A population of rabbits is observed to be 300 rabbits at time $t = 0$, where t is measured in days. After 2 weeks, the population doubled. Assuming exponential growth, how long would it take for the population to grow to $1{,}200$ rabbits?

Answer choices:

A 7 days

B 4 weeks

C 11 weeks 4 days

D 7 weeks 4 days

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Solution: B

After 2 weeks, the population doubled from 300 rabbits to 600 rabbits. Substituting these values into the exponential equation gives

$$600 = 300e^{k(14)}$$

$$\frac{600}{300} = e^{k(14)}$$

$$2 = e^{14k}$$

To solve for k, we'll apply the natural log to both sides of the equation, canceling the $\ln e$, and then rearrange.

$$\ln 2 = \ln e^{14k}$$

$$\ln 2 = 14k$$

$$k = \frac{\ln 2}{14}$$

Now that we have a value for the growth constant k, we can can figure out how long it'll take for the population to grow to $1{,}200$ rabbits.

$$1{,}200 = 300e^{\frac{\ln 2}{14}t}$$

$$4 = e^{\frac{\ln 2}{14}t}$$

Apply the natural log to both sides of the equation, canceling the $\ln e$, and then rearrange.

$$\ln 4 = \ln e^{\frac{\ln 2}{14}t}$$

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$$\ln 4 = \frac{\ln 2}{14}t$$

$$t = \frac{14 \ln 4}{\ln 2}$$

$$t \approx 28 \text{ days}$$

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